

energy weapons that are single-shot weapons, HPM weapons can operate any number of times as long as there is power source. This also implies reduced logistics cost and cost per shot.

HPM weapons' effects are scalable. Depending upon the magnitude of energy deposited on the target, the effects may be non-lethal or lethal. Active Denial System of the USA is a non-lethal HPM system designed for area denial, perimeter security and crowd control. It operates by heating the target's surface. In anti-personnel role, it is the skin of targeted human subjects. Also, HPM weapons have the potential of jamming the targeted system without any knowledge about the RF output of the system.

In the case of an HPM attack, targeted system recovery is extremely difficult as it may require component or sub-system-level troubleshooting. HPM weapons are highly effective against deeply-buried bunkers by targeting vulnerable electronic systems such as communications, power and air ventilation systems that support these bunkers.

Major limitations of HPM weapons include relatively shorter operational ranges, difficulty of damage assessment and likelihood of damage to friendly, unprotected systems.

HPM weapons are area weapons and therefore adversely affect all unprotected electronic systems within their lethal footprint. These could include civilian and friendly systems, which could be protected by using proper shielding. Proper planning before an HPM attack could prevent unintended damage to friendly assets.

In the case of an HPM weapon attack, target damage assessment is very difficult as there are no signs of any physical destruction. Just because a system has stopped operating or emitting does not mean it has been affected by an HPM at-

tack. Both absence and presence of emission from the targeted system does not reliably establish the efficacy of attack. However, advanced techniques may be used to measure second- and third-order effects to corroborate the results and enhance attack assessment.

HPM weapons have relatively shorter ranges as compared to high-energy laser weapons. In case of an HPM weapon system, range is proportional to both power output and antenna size. However, at sufficiently high microwave power levels, atmosphere at the antenna aperture becomes plasma, a phenomenon called atmospheric breakdown.

Plasma density increases with time duration of HPM pulses, eventually reaching the point where plasma reflects and absorbs RF energy. This not only renders the beam ineffective but also creates a shielding problem for the delivery platform.

Once atmospheric breakdown occurs, operational range can be increased only by increasing the antenna aperture, which may not be possible for portable or airborne designs.

For applications where portability is not a requirement and size is not a limiting factor, one can increase the effective aperture size and power on the target by using phase-locking techniques and combining multiple transmitters.

Another approach may be adopting designs that integrate antennae into the skin of the system, which allows aperture to be as large as the delivery platform. One such application scenario is that of point defence where the target is travelling towards the weapon system rather than weapon travelling towards the target. Ranets-E and Vigilant Eagle HPM systems are examples of such systems.

In the next part we will learn about e-bombs and representative HPM weapon systems.

To be continued...

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Meco Meters Private Limited

Plot No. EL-80, MIDC Electronic Zone,
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Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)

Fax : 0091-22-27673310, 27673330

Email : sales@mecoind.com

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